

Flood Impact Analysis Report

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Exploratory Data Analysis

Flood Frequency and Severity

The following plots show the distribution of flood frequency among respondents.

```
table(final_merged_data$flood_frequency)
```

```
2 periods a year, unpredictable how often
                                     5
                                     3.5
                                     2
                                     4
                                     1
                                     6
                                     1
Every time rain passes
                                     1
```

Severity

Every respondent experienced floodings to a severe extent

```
summary(final_merged_data$severity)
```

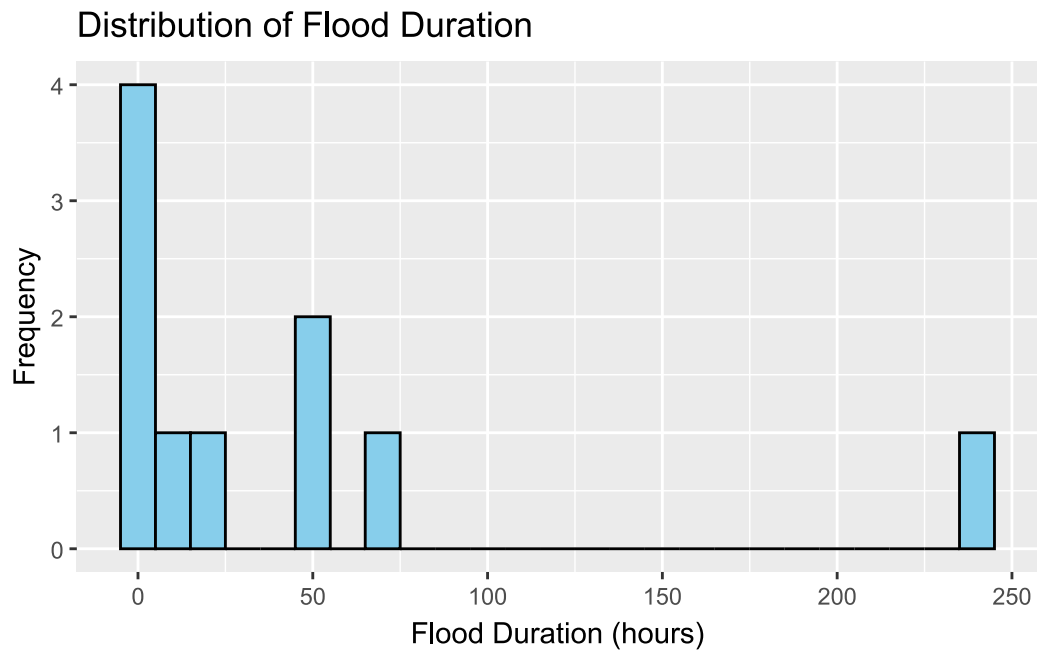
```
Severe
    10
```

Duration

The duration of the floods is shown in the following histogram, with one outlier removed (3600 hours extra)

```
final_merged_data <- final_merged_data %>% filter(duration_hours < 3600)
```

```
ggplot(final_merged_data, aes(x = duration_hours)) +
  geom_histogram(binwidth = 10, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Flood Duration",
       x = "Flood Duration (hours)",
       y = "Frequency")
```



Causes

The causes of flooding are mostly due to poor and blocked drainage

```
table(final_merged_data$causes)
```

Blocked drainage, proximity to lake	Poor drainage
1	4
Poor drainage, blocked drainage	Poor drainage, rubble
2	3

Making travelling impossible during these events

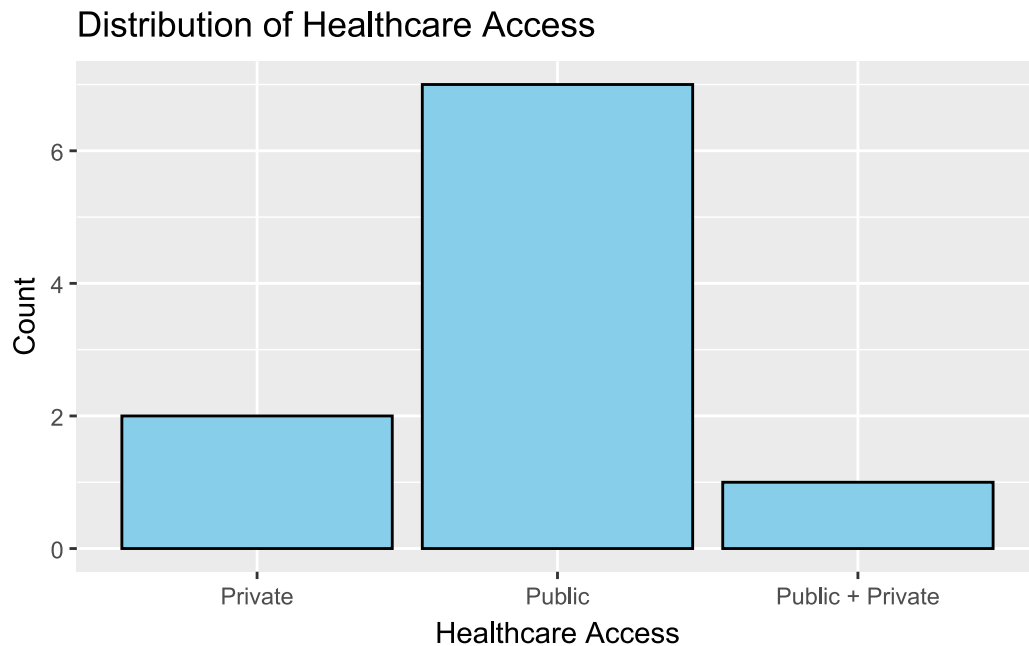
```
table(final_merged_data$travel_difficulty)
```

	Impossible	2
difficult	Impossible during, after 1 - 2 hours possible but still	1
possible	Impossible during, after 1 hour	1
impossible	Near	1
difficult	Very	3
Very difficult, children miss school, carry child to main road for further transport		1
recede	Very difficult, wait for water to	1

Healthcare access

Most people made use of the public facilities and some private. Private was preferred sometimes as public was not able to treat patients due to lack of medicines and other resources.

```
ggplot(final_merged_data, aes(x = public_or_private
)) +
  geom_bar(fill = "skyblue", color = "black") +
  labs(title = "Distribution of Healthcare Access",
        x = "Healthcare Access",
        y = "Count")
```



Distribution of Travel Time Difference

The distribution of travel time differences highlights the delays caused by flooding. Everyone thought it was more difficult to access healthcare during floodings. And people mostly walked with some using a boda (and one a taxi)

```
summary(final_merged_data$harder_to_access)
```

```
Mode      TRUE
logical    10
```

```
summary(final_merged_data$transport_type)
```

```
Length      Class      Mode
10 character character
```

```
# First removed an outlier
final_merged_data <- final_merged_data %>% filter(travel_time_diff < 300)

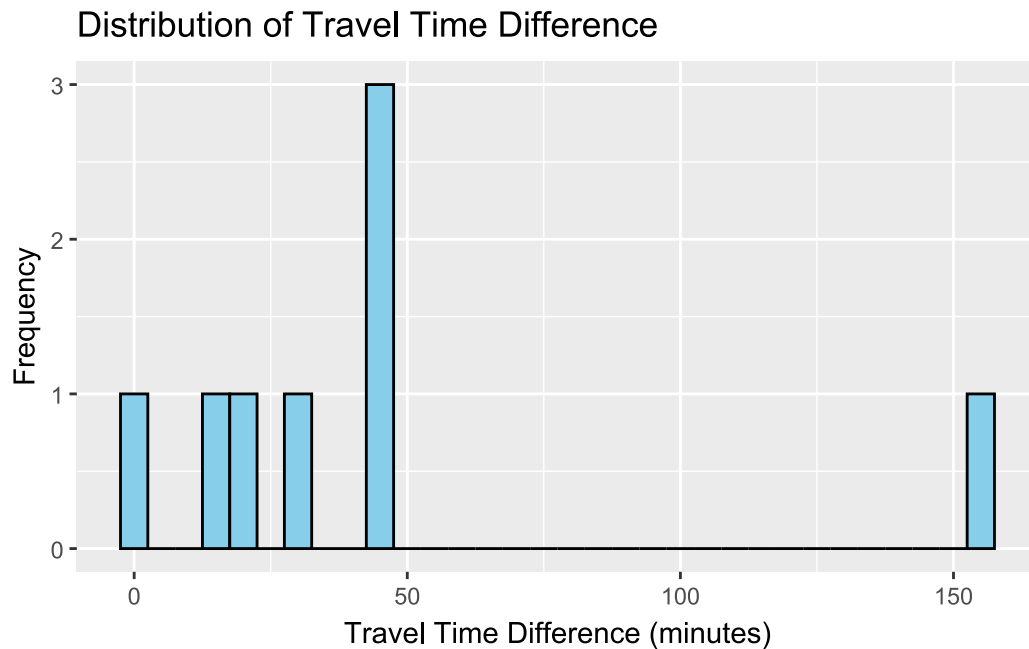
summary(final_merged_data[, c("normal_travel_time_min",
"flooding_travel_time_min", "travel_time_diff")])
```

normal_travel_time_min	flood_travel_time_min	travel_time_diff
Min. :25.00	Min. : 30.00	Min. : 0.00
1st Qu.:30.00	1st Qu.: 56.25	1st Qu.: 18.75
Median :45.00	Median : 82.50	Median : 37.50
Mean :41.88	Mean : 86.25	Mean : 44.38
3rd Qu.:47.50	3rd Qu.: 90.00	3rd Qu.: 45.00
Max. :60.00	Max. :210.00	Max. :155.00

```
summary(final_merged_data$water_height_cm)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
50.00	60.00	65.00	74.38	82.50	125.00

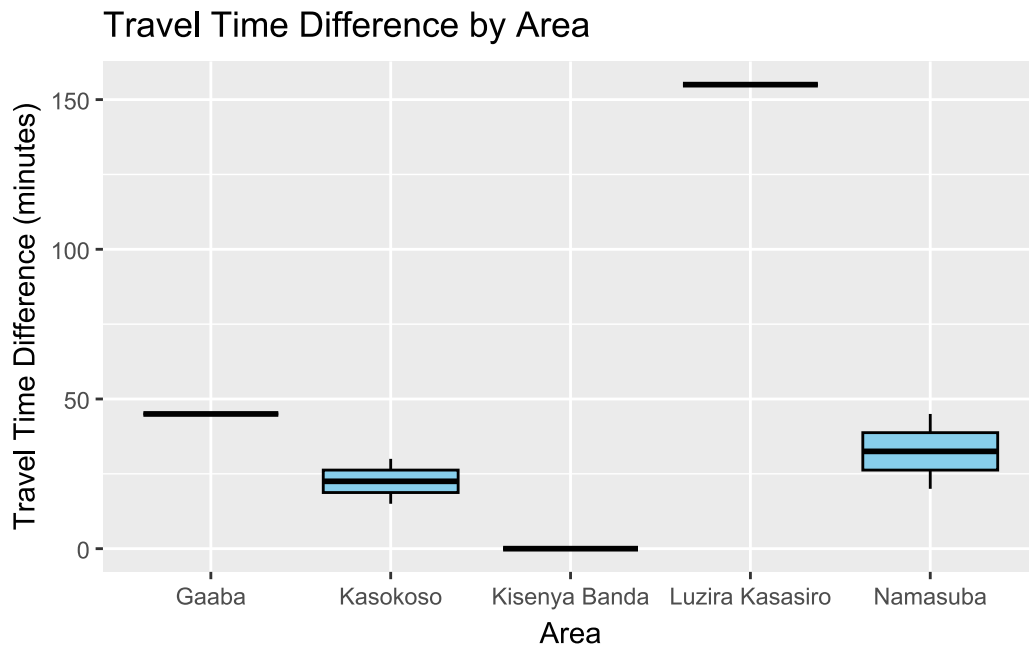
```
ggplot(final_merged_data, aes(x = travel_time_diff)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Travel Time Difference",
       x = "Travel Time Difference (minutes)",
       y = "Frequency")
```



Travel Time Difference by Area

Understanding the differences in travel time across areas provides insight into localized impacts:

```
ggplot(final_merged_data, aes(x = area_name, y = travel_time_diff)) +
  geom_boxplot(fill = "skyblue", color = "black") +
  labs(title = "Travel Time Difference by Area",
       x = "Area",
       y = "Travel Time Difference (minutes)")
```



Healthcare access was limited while everyone's experiences more health issues during floodings

```
summary(final_merged_data$health_problems)
```

```
Mode      TRUE
logical    8
```

```
summary(final_merged_data$problem_types)
```

```
Length    Class      Mode
      8 character character
```

Coping and suggestions

Most people coped by staying at home and building barriers and clearing rubbish. The suggestions were mostly about improving the drainage system, roads and removing rubbish from drainage.

Interpretation of Results

- **Flood Frequency:** Respondents experience frequent flooding, with most experiencing it multiple times a year.
 - **Health impacts:** Many experience health problems during floodings.
 - **Travel Time Differences:** The histogram shows that flooding significantly increases travel times for most respondents, with some extreme delays observed.
 - **Area Variability:** Boxplots reveal that certain areas experience greater travel time delays, which could be due to infrastructure issues or geographic challenges.
 - **Healthcare Access:** Respondents primarily rely on public healthcare facilities, but some opt for private care when public facilities are unable to provide treatment.
 - **Coping Mechanisms:** Respondents employ a variety of coping strategies during floods, such as staying at home and clearing debris. These strategies can help mitigate the impacts of flooding.
 - **Suggestions for Improvement:** Respondents suggest improving drainage systems, roads, and waste management to reduce the frequency and severity of floods.
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